

```
import pandas as pd
```

```
df=pd.read_csv(r'C:\Users\HP\Downloads\flights.csv\flights.csv')
```

```
df.head(10)
```

	id	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time
\								
0	0	2013	1	1	517.0	515	2.0	830.0
1	1	2013	1	1	533.0	529	4.0	850.0
2	2	2013	1	1	542.0	540	2.0	923.0
3	3	2013	1	1	544.0	545	-1.0	1004.0
4	4	2013	1	1	554.0	600	-6.0	812.0
5	5	2013	1	1	554.0	558	-4.0	740.0
6	6	2013	1	1	555.0	600	-5.0	913.0
7	7	2013	1	1	557.0	600	-3.0	709.0
8	8	2013	1	1	557.0	600	-3.0	838.0
9	9	2013	1	1	558.0	600	-2.0	753.0

	sched_arr_time	arr_delay	...	flight	tailnum	origin	dest	air_time
\								
0	819	11.0	...	1545	N14228	EWB	IAH	227.0
1	830	20.0	...	1714	N24211	LGA	IAH	227.0
2	850	33.0	...	1141	N619AA	JFK	MIA	160.0
3	1022	-18.0	...	725	N804JB	JFK	BQN	183.0
4	837	-25.0	...	461	N668DN	LGA	ATL	116.0
5	728	12.0	...	1696	N39463	EWB	ORD	150.0
6	854	19.0	...	507	N516JB	EWB	FLL	158.0
7	723	-14.0	...	5708	N829AS	LGA	IAD	53.0
8	846	-8.0	...	79	N593JB	JFK	MCO	140.0
9	745	8.0	...	301	N3ALAA	LGA	ORD	138.0

	distance	hour	minute	time_hour			
0	1400	5	15	2013-01-01	05:00:00		United Air Lines
1	1416	5	29	2013-01-01	05:00:00		United Air Lines
2	1089	5	40	2013-01-01	05:00:00		American Airlines
3	1576	5	45	2013-01-01	05:00:00		JetBlue
4	762	6	0	2013-01-01	06:00:00		Delta Air Lines
5	719	5	58	2013-01-01	05:00:00		United Air Lines
6	1065	6	0	2013-01-01	06:00:00		JetBlue
7	229	6	0	2013-01-01	06:00:00		ExpressJet Airlines
8	944	6	0	2013-01-01	06:00:00		JetBlue
9	733	6	0	2013-01-01	06:00:00		American Airlines

[10 rows x 21 columns]

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 336776 entries, 0 to 336775
Data columns (total 21 columns):
```

#	Column	Non-Null Count	Dtype
0	id	336776 non-null	int64
1	year	336776 non-null	int64
2	month	336776 non-null	int64
3	day	336776 non-null	int64
4	dep_time	328521 non-null	float64
5	sched_dep_time	336776 non-null	int64
6	dep_delay	328521 non-null	float64
7	arr_time	328063 non-null	float64
8	sched_arr_time	336776 non-null	int64
9	arr_delay	327346 non-null	float64
10	carrier	336776 non-null	object
11	flight	336776 non-null	int64
12	tailnum	334264 non-null	object
13	origin	336776 non-null	object
14	dest	336776 non-null	object
15	air_time	327346 non-null	float64
16	distance	336776 non-null	int64
17	hour	336776 non-null	int64

```
18 minute          336776 non-null int64
19 time_hour       336776 non-null object
20 name            336776 non-null object
dtypes: float64(5), int64(10), object(6)
memory usage: 54.0+ MB
```

```
df.isnull().sum()
```

```
id          0
year        0
month       0
day         0
dep_time    8255
sched_dep_time  0
dep_delay   8255
arr_time    8713
sched_arr_time  0
arr_delay   9430
carrier     0
flight      0
tailnum     2512
origin      0
dest        0
air_time    9430
distance    0
hour        0
minute      0
time_hour   0
name        0
dtype: int64
```

```
df.name.value_counts()
```

```
name
United Air Lines Inc.      58665
JetBlue Airways            54635
ExpressJet Airlines Inc.   54173
Delta Air Lines Inc.       48110
American Airlines Inc.     32729
Envoy Air                  26397
US Airways Inc.            20536
Endeavor Air Inc.          18460
Southwest Airlines Co.     12275
Virgin America              5162
AirTran Airways Corporation 3260
Alaska Airlines Inc.        714
Frontier Airlines Inc.      685
Mesa Airlines Inc.          601
Hawaiian Airlines Inc.     342
```

SkyWest Airlines Inc. 32

Name: count, dtype: int64

df.carrier.value_counts()

```
carrier
UA      58665
B6      54635
EV      54173
DL      48110
AA      32729
MQ      26397
US      20536
9E      18460
WN      12275
VX       5162
FL       3260
AS        714
F9        685
YV        601
HA        342
OO         32
```

Name: count, dtype: int64

df.origin.value_counts()

```
origin
EWR     120835
JFK     111279
LGA     104662
```

Name: count, dtype: int64

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.kdeplot(df.loc[(df['origin']=='EWR'),'id'],color='navy',shade=True)
sns.kdeplot(df.loc[(df['origin']=='JFK'),'id'],color='green')
sns.kdeplot(df.loc[(df['origin']=='LGA'),'id'],color='purple')
plt.xlabel('No. of flights')
plt.ylabel('Probability Density')
plt.show()
```

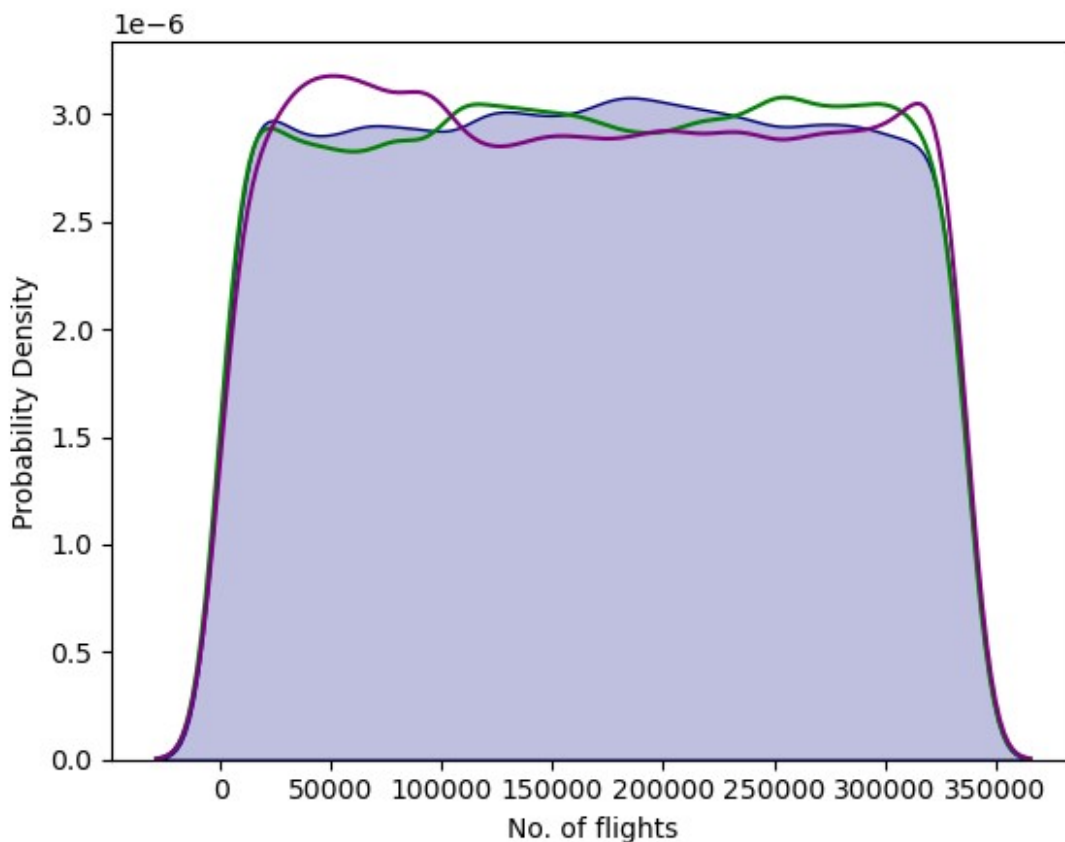
C:\Users\HP\AppData\Local\Temp\ipykernel_2720\2323897703.py:1:

FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

```
sns.kdeplot(df.loc[(df['origin']=='EWR'),'id'],color='navy',shade=True
```

```
)
D:\Users\HP\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed
in a future version. Convert inf values to NaN before operating
instead.
  with pd.option_context('mode.use_inf_as_na', True):
D:\Users\HP\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed
in a future version. Convert inf values to NaN before operating
instead.
  with pd.option_context('mode.use_inf_as_na', True):
D:\Users\HP\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed
in a future version. Convert inf values to NaN before operating
instead.
  with pd.option_context('mode.use_inf_as_na', True):
```



```
df.dep_time.mean()
1349.1099473093045

ls=['dep_time', 'sched_dep_time', 'dep_delay']
df[ls]
```

	dep_time	sched_dep_time	dep_delay
0	517.0	515	2.0
1	533.0	529	4.0
2	542.0	540	2.0
3	544.0	545	-1.0
4	554.0	600	-6.0
...	...	...	...
336771	NaN	1455	NaN
336772	NaN	2200	NaN
336773	NaN	1210	NaN
336774	NaN	1159	NaN
336775	NaN	840	NaN

[336776 rows x 3 columns]

```
df.dep_delay.median()
```

-2.0

```
df.dep_delay=df.dep_delay.fillna(df.dep_delay.median())
```

```
df[ls]
```

	dep_time	sched_dep_time	dep_delay
0	517.0	515	2.0
1	533.0	529	4.0
2	542.0	540	2.0
3	544.0	545	-1.0
4	554.0	600	-6.0
...	...	...	...
336771	NaN	1455	-2.0
336772	NaN	2200	-2.0
336773	NaN	1210	-2.0
336774	NaN	1159	-2.0
336775	NaN	840	-2.0

[336776 rows x 3 columns]

```
df.dep_time=df.dep_time.fillna(df.sched_dep_time+df.dep_delay)
```

```
df[ls]
```

	dep_time	sched_dep_time	dep_delay
0	517.0	515	2.0
1	533.0	529	4.0
2	542.0	540	2.0
3	544.0	545	-1.0
4	554.0	600	-6.0
...	...	...	...
336771	1453.0	1455	-2.0
336772	2198.0	2200	-2.0

336773	1208.0	1210	-2.0
336774	1157.0	1159	-2.0
336775	838.0	840	-2.0

[336776 rows x 3 columns]

```
df.isnull().sum()
```

```
id          0
year        0
month       0
day         0
dep_time    0
sched_dep_time  0
dep_delay   0
arr_time    8713
sched_arr_time  0
arr_delay   9430
carrier     0
flight      0
tailnum    2512
origin      0
dest        0
air_time    9430
distance    0
hour        0
minute      0
time_hour   0
name        0
```

```
dtype: int64
```

```
ls1=['arr_time','sched_arr_time','arr_delay']
df[ls1]
```

	arr_time	sched_arr_time	arr_delay
0	830.0	819	11.0
1	850.0	830	20.0
2	923.0	850	33.0
3	1004.0	1022	-18.0
4	812.0	837	-25.0
...	...	...	...
336771	NaN	1634	NaN
336772	NaN	2312	NaN
336773	NaN	1330	NaN
336774	NaN	1344	NaN
336775	NaN	1020	NaN

[336776 rows x 3 columns]

```
df.arr_delay.mean().round()
```

7.0

```
df.arr_delay=df.arr_delay.fillna(7)
```

```
df[ls1]
```

	arr_time	sched_arr_time	arr_delay
0	830.0	819	11.0
1	850.0	830	20.0
2	923.0	850	33.0
3	1004.0	1022	-18.0
4	812.0	837	-25.0
...	...	...	...
336771	NaN	1634	7.0
336772	NaN	2312	7.0
336773	NaN	1330	7.0
336774	NaN	1344	7.0
336775	NaN	1020	7.0

[336776 rows x 3 columns]

```
df.arr_time=df.arr_time.fillna(df.sched_arr_time+df.arr_delay)
```

```
df[ls1]
```

	arr_time	sched_arr_time	arr_delay
0	830.0	819	11.0
1	850.0	830	20.0
2	923.0	850	33.0
3	1004.0	1022	-18.0
4	812.0	837	-25.0
...	...	...	...
336771	1641.0	1634	7.0
336772	2319.0	2312	7.0
336773	1337.0	1330	7.0
336774	1351.0	1344	7.0
336775	1027.0	1020	7.0

[336776 rows x 3 columns]

```
df.arr_time=df.arr_time.round()
```

```
df[ls1]
```

	arr_time	sched_arr_time	arr_delay
0	830.0	819	11.0
1	850.0	830	20.0
2	923.0	850	33.0
3	1004.0	1022	-18.0
4	812.0	837	-25.0
...	...	...	...

336771	1641.0	1634	7.0
336772	2319.0	2312	7.0
336773	1337.0	1330	7.0
336774	1351.0	1344	7.0
336775	1027.0	1020	7.0

[336776 rows x 3 columns]

df.head()

	id	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time
\								
0	0	2013	1	1	517.0	515	2.0	830.0
1	1	2013	1	1	533.0	529	4.0	850.0
2	2	2013	1	1	542.0	540	2.0	923.0
3	3	2013	1	1	544.0	545	-1.0	1004.0
4	4	2013	1	1	554.0	600	-6.0	812.0

	sched_arr_time	arr_delay	...	flight	tailnum	origin	dest	air_time
\								
0	819	11.0	...	1545	N14228	EWB	IAH	227.0
1	830	20.0	...	1714	N24211	LGA	IAH	227.0
2	850	33.0	...	1141	N619AA	JFK	MIA	160.0
3	1022	-18.0	...	725	N804JB	JFK	BQN	183.0
4	837	-25.0	...	461	N668DN	LGA	ATL	116.0

	distance	hour	minute	time_hour	name
0	1400	5	15	2013-01-01 05:00:00	United Air Lines Inc.
1	1416	5	29	2013-01-01 05:00:00	United Air Lines Inc.
2	1089	5	40	2013-01-01 05:00:00	American Airlines Inc.
3	1576	5	45	2013-01-01 05:00:00	JetBlue Airways
4	762	6	0	2013-01-01 06:00:00	Delta Air Lines Inc.

[5 rows x 21 columns]

df.air_time.median()

129.0

```
df.air_time=df.air_time.fillna(df.air_time.median())
```

```
df.isnull().sum()
```

```
id          0
year        0
month       0
day         0
dep_time    0
sched_dep_time  0
dep_delay   0
arr_time    0
sched_arr_time  0
arr_delay   0
carrier     0
flight      0
tailnum     2512
origin      0
dest        0
air_time    0
distance    0
hour        0
minute      0
time_hour   0
name        0
dtype: int64
```

```
df.month.value_counts()
```

```
month
7      29425
8      29327
10     28889
3      28834
5      28796
4      28330
6      28243
12     28135
9      27574
11     27268
1      27004
2      24951
Name: count, dtype: int64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 336776 entries, 0 to 336775
Data columns (total 21 columns):
```

#	Column	Non-Null	Count	Dtype
0	id	336776	non-null	int64
1	year	336776	non-null	int64
2	month	336776	non-null	int64
3	day	336776	non-null	int64
4	dep_time	336776	non-null	float64
5	sched_dep_time	336776	non-null	int64
6	dep_delay	336776	non-null	float64
7	arr_time	336776	non-null	float64
8	sched_arr_time	336776	non-null	int64
9	arr_delay	336776	non-null	float64
10	carrier	336776	non-null	object
11	flight	336776	non-null	int64
12	tailnum	334264	non-null	object
13	origin	336776	non-null	object
14	dest	336776	non-null	object
15	air_time	336776	non-null	float64
16	distance	336776	non-null	int64
17	hour	336776	non-null	int64
18	minute	336776	non-null	int64
19	time_hour	336776	non-null	object
20	name	336776	non-null	object

dtypes: float64(5), int64(10), object(6)

memory usage: 54.0+ MB

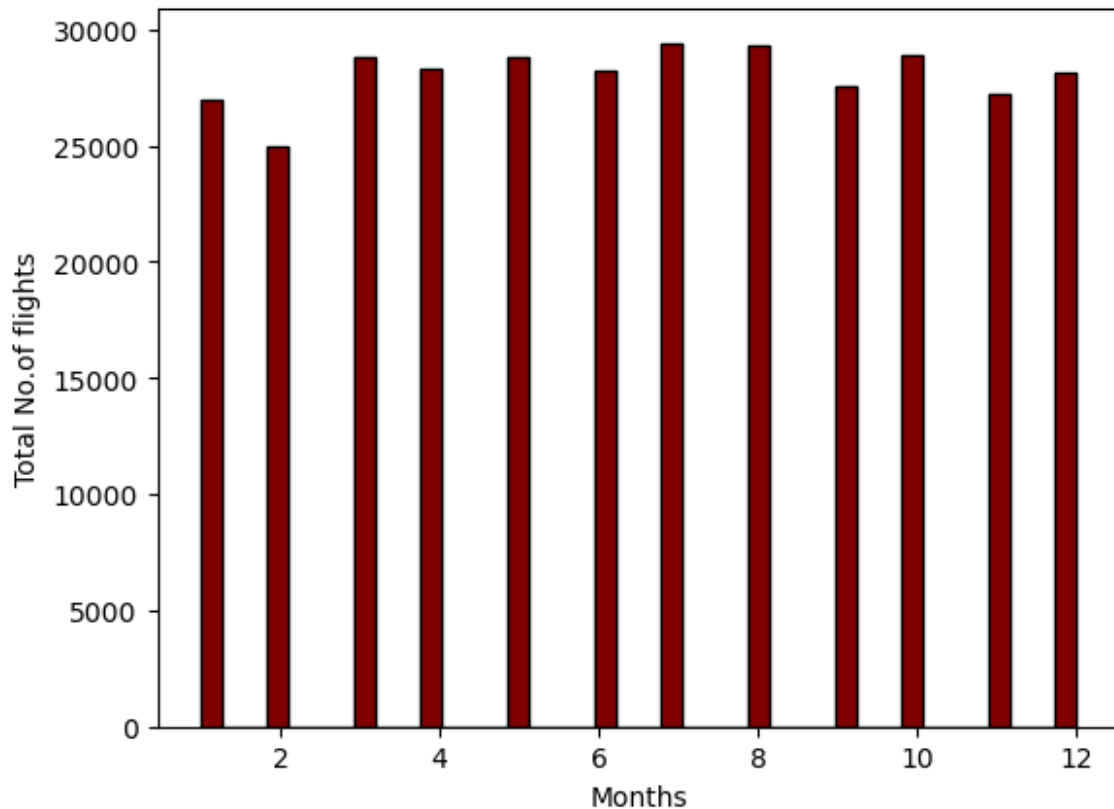
```
data=df.month
```

```
plt.hist(data,bins=40,color='maroon',edgecolor='black',rwidth=1.5)
```

```
plt.xlabel('Months')
```

```
plt.ylabel('Total No.of flights')
```

```
plt.show()
```



```
lr=['air_time','distance','origin']
df[lr].head()
```

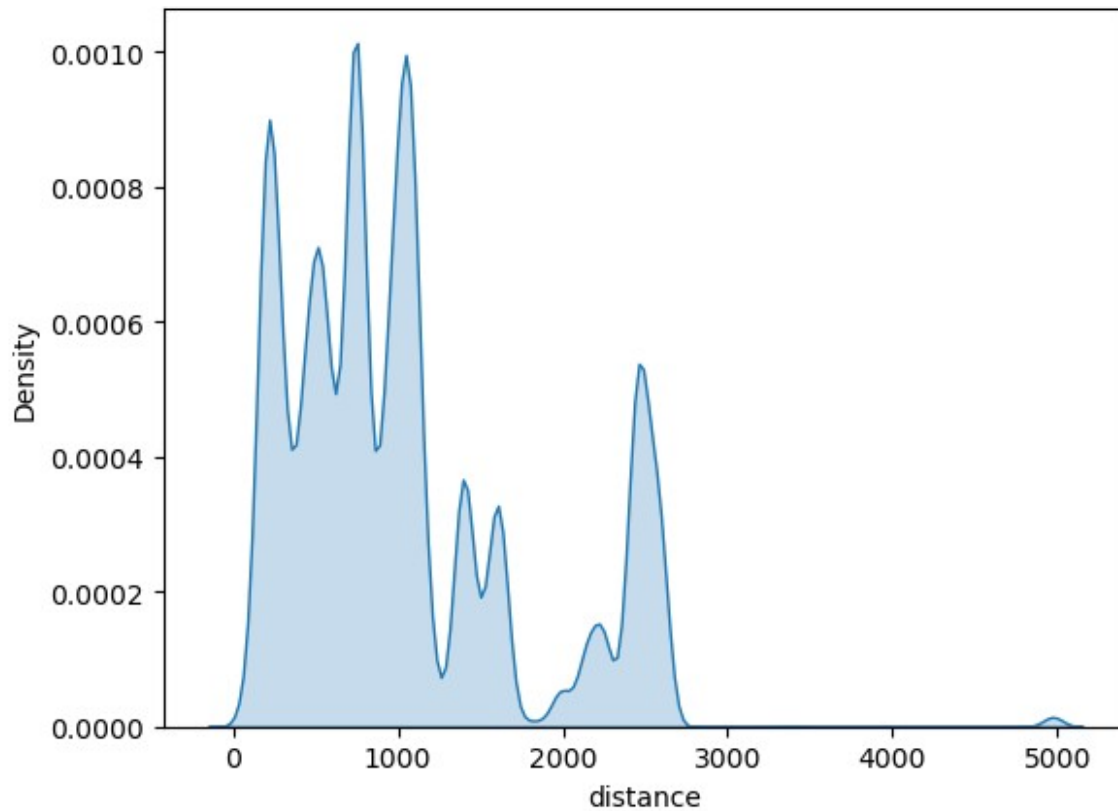
	air_time	distance	origin
0	227.0	1400	EWB
1	227.0	1416	LGA
2	160.0	1089	JFK
3	183.0	1576	JFK
4	116.0	762	LGA

```
x=df.distance
y=df.air_time
sns.kdeplot(x,fill=True)
```

```
D:\Users\HP\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed
in a future version. Convert inf values to NaN before operating
instead.
```

```
with pd.option_context('mode.use_inf_as_na', True):
```

```
<Axes: xlabel='distance', ylabel='Density'>
```



```
df.tailnum
```

```
0      N14228
1      N24211
2      N619AA
3      N804JB
4      N668DN
```

```
...
```

```
336771    NaN
336772    NaN
336773    N535MQ
336774    N511MQ
336775    N839MQ
```

```
Name: tailnum, Length: 336776, dtype: object
```

```
df.head()
```

	id	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time
0	0	2013	1	1	517.0	515	2.0	830.0
1	1	2013	1	1	533.0	529	4.0	850.0
2	2	2013	1	1	542.0	540	2.0	923.0

3	3	2013	1	1	544.0	545	-1.0	1004.0
4	4	2013	1	1	554.0	600	-6.0	812.0

	sched_arr_time	arr_delay	...	flight	tailnum	origin	dest	air_time
0	819	11.0	...	1545	N14228	EWB	IAH	227.0
1	830	20.0	...	1714	N24211	LGA	IAH	227.0
2	850	33.0	...	1141	N619AA	JFK	MIA	160.0
3	1022	-18.0	...	725	N804JB	JFK	BQN	183.0
4	837	-25.0	...	461	N668DN	LGA	ATL	116.0

	distance	hour	minute	time_hour	name
0	1400	5	15	2013-01-01 05:00:00	United Air Lines Inc.
1	1416	5	29	2013-01-01 05:00:00	United Air Lines Inc.
2	1089	5	40	2013-01-01 05:00:00	American Airlines Inc.
3	1576	5	45	2013-01-01 05:00:00	JetBlue Airways
4	762	6	0	2013-01-01 06:00:00	Delta Air Lines Inc.

[5 rows x 21 columns]

df.dest.value_counts()

```
dest
ORD    17283
ATL    17215
LAX    16174
BOS    15508
MCO    14082
...
MTJ      15
SBN      10
ANC       8
LEX       1
LGA       1
Name: count, Length: 105, dtype: int64
```

```
lr1=['origin','dest']
df[lr1]
```

	origin	dest
0	EWK	IAH
1	LGA	IAH
2	JFK	MIA
3	JFK	BQN
4	LGA	ATL
...	...	...
336771	JFK	DCA
336772	LGA	SYR
336773	LGA	BNA
336774	LGA	CLE
336775	LGA	RDU

[336776 rows x 2 columns]

```
df[df['origin']=='EWK']['dest'].value_counts()
```

dest	
ORD	6100
BOS	5327
SFO	5127
CLT	5026
ATL	5022
...	
HDN	15
MTJ	15
ANC	8
SBN	4
LGA	1

Name: count, Length: 86, dtype: int64

```
df[df['origin']=='LGA']['dest'].value_counts()
```

dest	
ATL	10263
ORD	8857
CLT	6168
MIA	5781
DTW	5040
...	
CAE	12
AVL	10
SBN	6
MYR	3
LEX	1

Name: count, Length: 68, dtype: int64

```
EWK=df[df['origin']=='EWK']['dest'].value_counts()
LGA=df[df['origin']=='LGA']['dest'].value_counts()
JFK=df[df['origin']=='JFK']['dest'].value_counts()
```

```
df1=pd.DataFrame([EWR,LGA,JFK])  
df1.index=['EWR','LGA','JFK']  
df1.plot(kind='bar',stacked=True,figsize=(10,5))
```

<Axes: >

dest	
ORD	
BOS	
SFO	
CLT	
ATL	
MCO	
LAX	
IAH	
FLL	
DTW	
DFW	
DEN	
PHX	
CVG	
MIA	
STL	
MSP	
PBI	
BNA	
TPA	
MDW	
LAS	
SEA	
CLE	
DCA	
RIC	
RDU	
RSW	
CHS	
MCI	
IND	
JAX	
IAD	
MSY	
DAY	
SAN	
GSO	
MKE	
SJU	
HOU	
BUF	
AUS	
BTV	
MEM	
SDF	
MHT	
SNA	
PWM	
OMA	
GSP	
SAV	
CMH	
GRR	
ORF	
PDX	

```
df.head()
```

	id	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time
0	0	2013	1	1	517.0	515	2.0	830.0
1	1	2013	1	1	533.0	529	4.0	850.0
2	2	2013	1	1	542.0	540	2.0	923.0
3	3	2013	1	1	544.0	545	-1.0	1004.0
4	4	2013	1	1	554.0	600	-6.0	812.0

	sched_arr_time	arr_delay	...	flight	tailnum	origin	dest	air_time
0	819	11.0	...	1545	N14228	EWB	IAH	227.0
1	830	20.0	...	1714	N24211	LGA	IAH	227.0
2	850	33.0	...	1141	N619AA	JFK	MIA	160.0
3	1022	-18.0	...	725	N804JB	JFK	BQN	183.0
4	837	-25.0	...	461	N668DN	LGA	ATL	116.0

	distance	hour	minute	time_hour	name
0	1400	5	15	2013-01-01 05:00:00	United Air Lines Inc.
1	1416	5	29	2013-01-01 05:00:00	United Air Lines Inc.
2	1089	5	40	2013-01-01 05:00:00	American Airlines Inc.
3	1576	5	45	2013-01-01 05:00:00	JetBlue Airways
4	762	6	0	2013-01-01 06:00:00	Delta Air Lines Inc.

```
[5 rows x 21 columns]
```

```
df.name.value_counts()
```

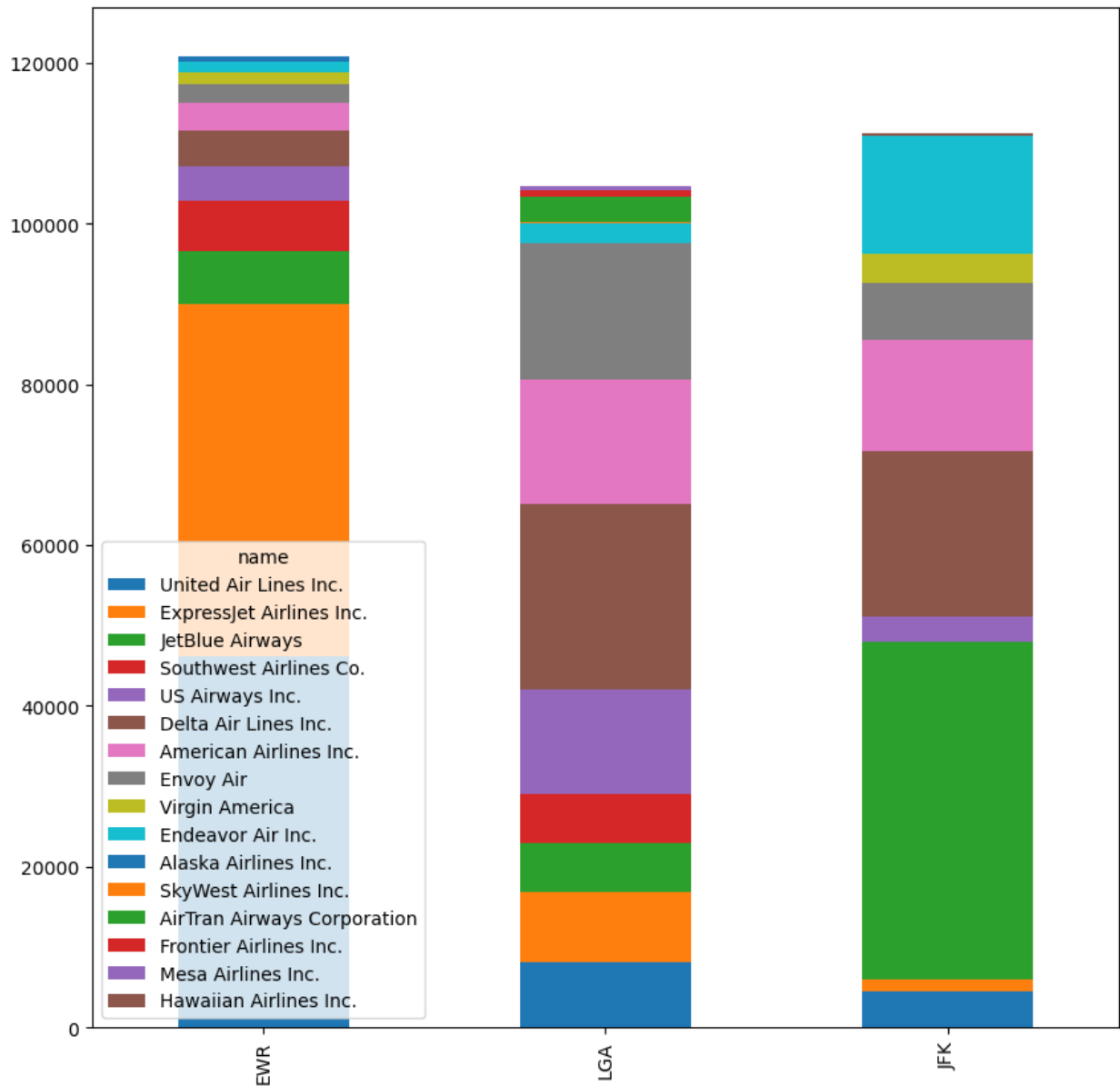
name	
United Air Lines Inc.	58665
JetBlue Airways	54635
ExpressJet Airlines Inc.	54173
Delta Air Lines Inc.	48110
American Airlines Inc.	32729
Envoy Air	26397

US Airways Inc.	20536
Endeavor Air Inc.	18460
Southwest Airlines Co.	12275
Virgin America	5162
AirTran Airways Corporation	3260
Alaska Airlines Inc.	714
Frontier Airlines Inc.	685
Mesa Airlines Inc.	601
Hawaiian Airlines Inc.	342
SkyWest Airlines Inc.	32

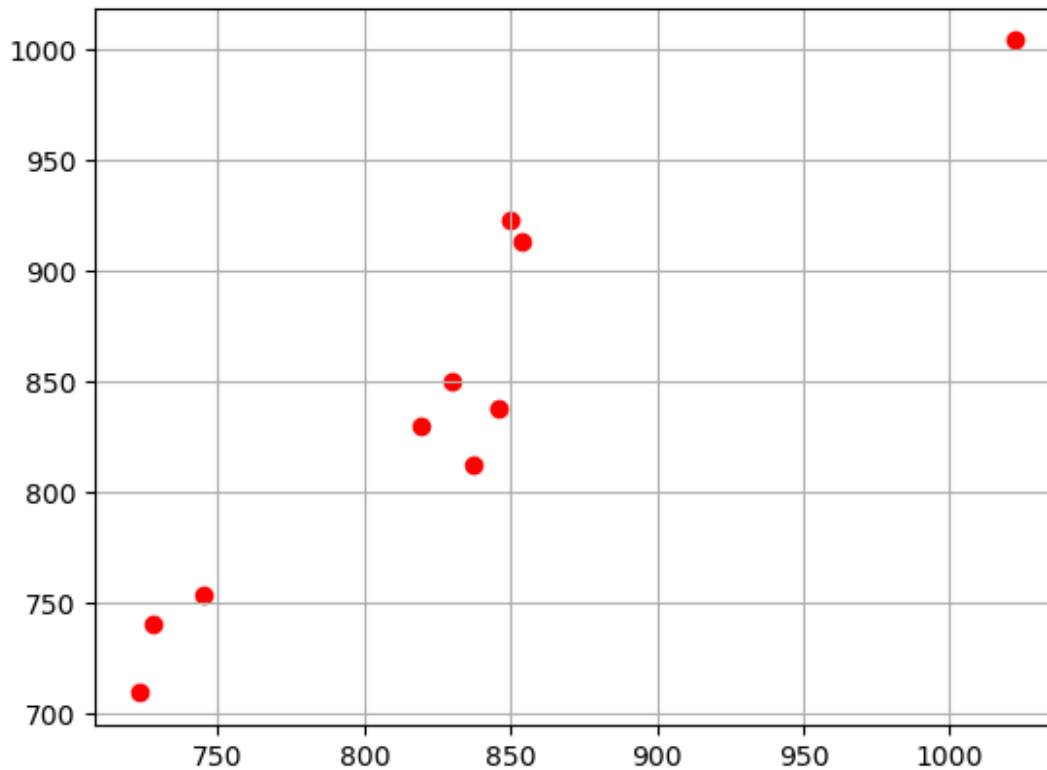
Name: count, dtype: int64

```
EWB=df[df['origin']=='EWB']['name'].value_counts()  
LGA=df[df['origin']=='LGA']['name'].value_counts()  
JFK=df[df['origin']=='JFK']['name'].value_counts()  
df1=pd.DataFrame([EWB,LGA,JFK])  
df1.index=['EWB','LGA','JFK']  
df1.plot(kind='bar',stacked=True,figsize=(10,10))
```

<Axes: >



```
x=df.sched_arr_time.head(10)
y=df.arr_time.head(10)
plt.scatter(x,y,color='red')
plt.grid(True)
plt.show()
```



```
df.carrier.value_counts()
```

```
carrier
```

```
UA      58665
```

```
B6      54635
```

```
EV      54173
```

```
DL      48110
```

```
AA      32729
```

```
MQ      26397
```

```
US      20536
```

```
9E      18460
```

```
WN      12275
```

```
VX       5162
```

```
FL       3260
```

```
AS        714
```

```
F9        685
```

```
YV        601
```

```
HA        342
```

```
00         32
```

```
Name: count, dtype: int64
```

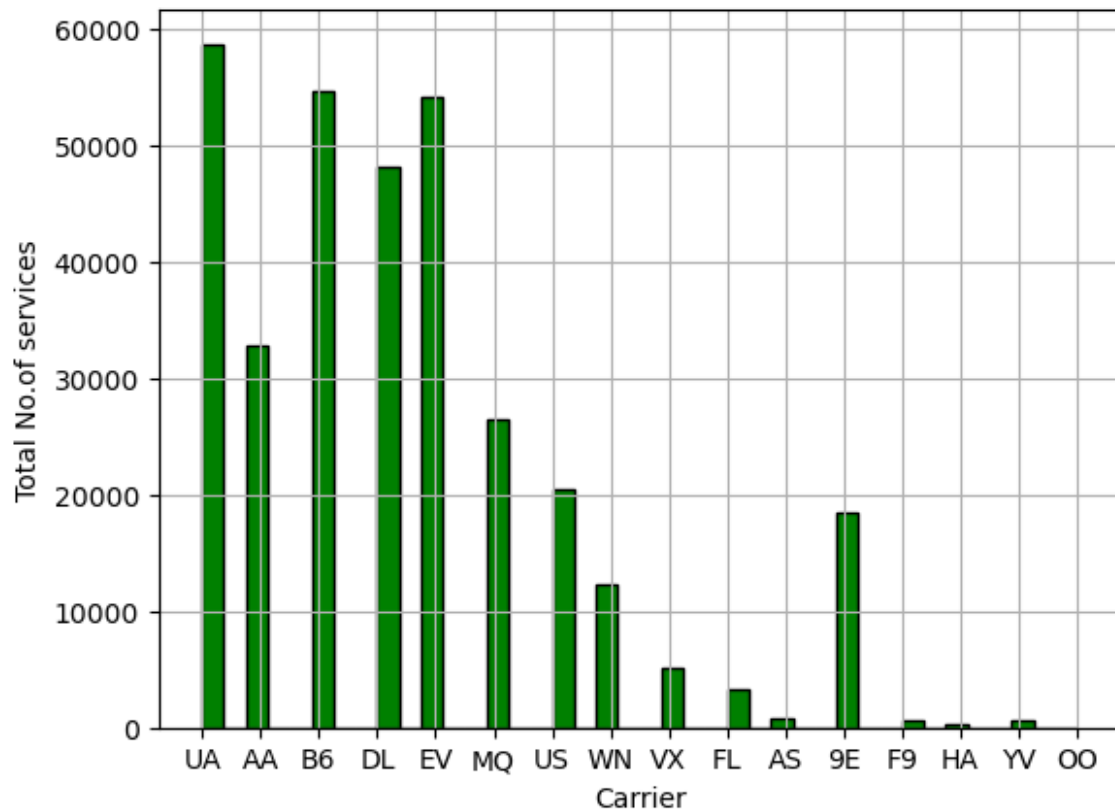
```
data=df.carrier
```

```
plt.hist(data,bins=40,color='green',edgecolor='black',rwidth=1.5)
```

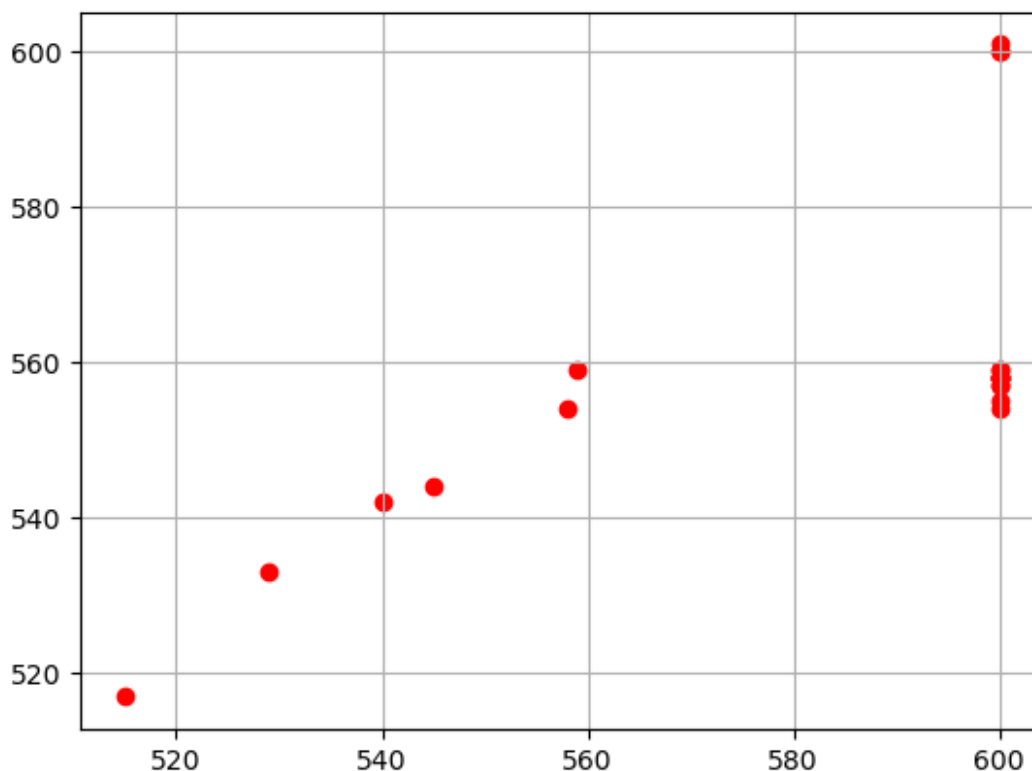
```
plt.xlabel('Carrier')
```

```
plt.ylabel('Total No.of services')
```

```
plt.grid(True)
plt.show()
```



```
x=df.sched_dep_time.head(20)
y=df.dep_time.head(20)
plt.scatter(x,y,color='red')
plt.grid(True)
plt.show()
```



```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 336776 entries, 0 to 336775
```

```
Data columns (total 21 columns):
```

#	Column	Non-Null Count	Dtype
0	id	336776 non-null	int64
1	year	336776 non-null	int64
2	month	336776 non-null	int64
3	day	336776 non-null	int64
4	dep_time	336776 non-null	float64
5	sched_dep_time	336776 non-null	int64
6	dep_delay	336776 non-null	float64
7	arr_time	336776 non-null	float64
8	sched_arr_time	336776 non-null	int64
9	arr_delay	336776 non-null	float64
10	carrier	336776 non-null	object
11	flight	336776 non-null	int64
12	tailnum	334264 non-null	object
13	origin	336776 non-null	object
14	dest	336776 non-null	object
15	air_time	336776 non-null	float64
16	distance	336776 non-null	int64
17	hour	336776 non-null	int64

```

18 minute          336776 non-null int64
19 time_hour       336776 non-null object
20 name            336776 non-null object
dtypes: float64(5), int64(10), object(6)
memory usage: 54.0+ MB

```

```
df = df.drop(['id', 'tailnum'], axis=1)
```

```
df.head(10)
```

	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	\
0	2013	1	1	517.0	515	2.0	830.0	
1	2013	1	1	533.0	529	4.0	850.0	
2	2013	1	1	542.0	540	2.0	923.0	
3	2013	1	1	544.0	545	-1.0	1004.0	
4	2013	1	1	554.0	600	-6.0	812.0	
5	2013	1	1	554.0	558	-4.0	740.0	
6	2013	1	1	555.0	600	-5.0	913.0	
7	2013	1	1	557.0	600	-3.0	709.0	
8	2013	1	1	557.0	600	-3.0	838.0	
9	2013	1	1	558.0	600	-2.0	753.0	

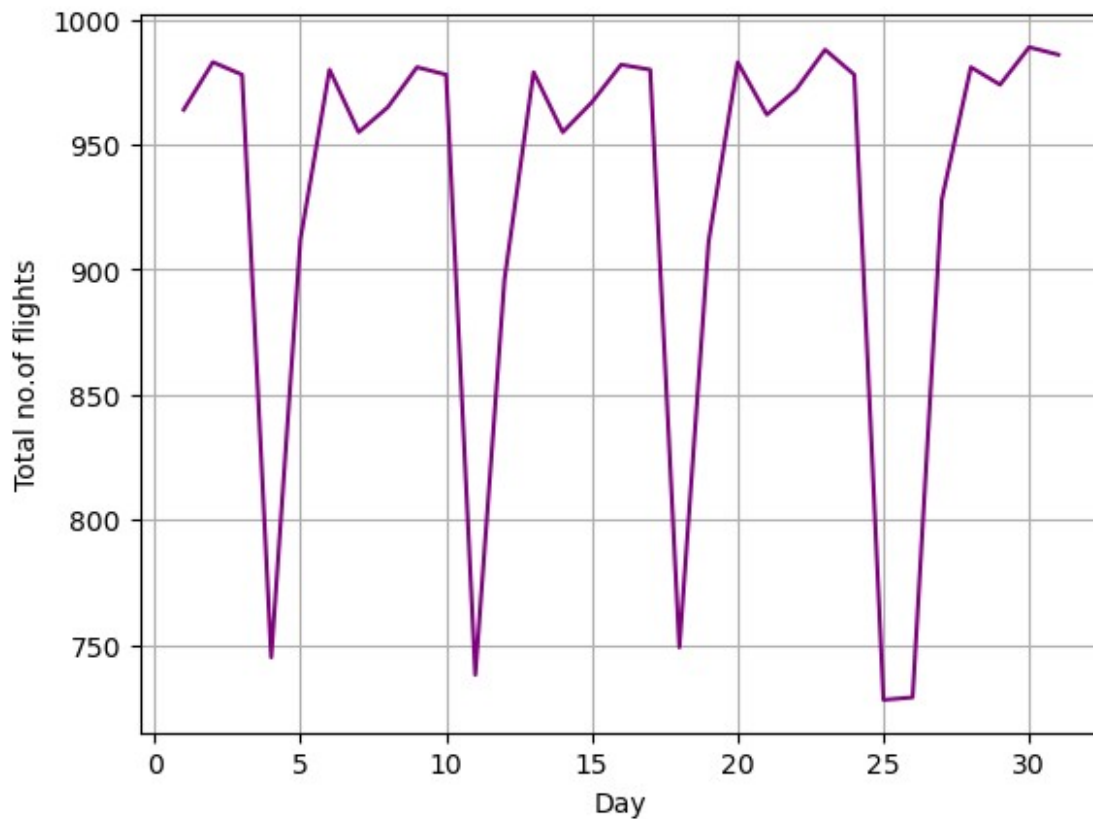
	sched_arr_time	arr_delay	carrier	flight	origin	dest	air_time	\
0		819	11.0	UA	1545	EWR IAH	227.0	
1400								
1		830	20.0	UA	1714	LGA IAH	227.0	
1416								
2		850	33.0	AA	1141	JFK MIA	160.0	
1089								
3		1022	-18.0	B6	725	JFK BQN	183.0	
1576								
4		837	-25.0	DL	461	LGA ATL	116.0	
762								
5		728	12.0	UA	1696	EWR ORD	150.0	
719								
6		854	19.0	B6	507	EWR FLL	158.0	
1065								
7		723	-14.0	EV	5708	LGA IAD	53.0	
229								
8		846	-8.0	B6	79	JFK MCO	140.0	
944								
9		745	8.0	AA	301	LGA ORD	138.0	
733								

	hour	minute	time_hour	name
0	5	15	2013-01-01 05:00:00	United Air Lines Inc.
1	5	29	2013-01-01 05:00:00	United Air Lines Inc.
2	5	40	2013-01-01 05:00:00	American Airlines Inc.
3	5	45	2013-01-01 05:00:00	JetBlue Airways

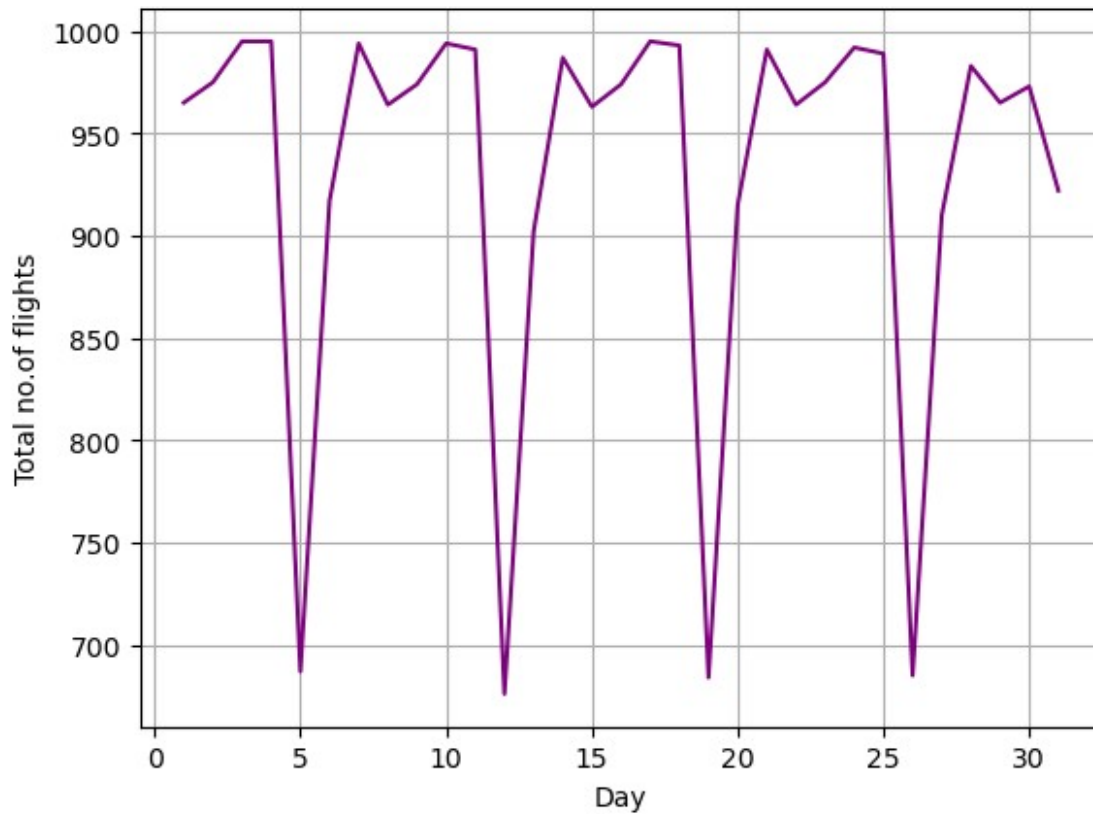
4	6	0	2013-01-01 06:00:00	Delta Air Lines Inc.
5	5	58	2013-01-01 05:00:00	United Air Lines Inc.
6	6	0	2013-01-01 06:00:00	JetBlue Airways
7	6	0	2013-01-01 06:00:00	ExpressJet Airlines Inc.
8	6	0	2013-01-01 06:00:00	JetBlue Airways
9	6	0	2013-01-01 06:00:00	American Airlines Inc.

```
def per_month(n):
    x = df.day.unique()
    y = df[df['month']==n]['day'].value_counts().sort_index()
    plt.plot(x,y, color='purple')
    plt.grid(True)
    plt.xlabel('Day')
    plt.ylabel('Total no.of flights')
    plt.show()
```

```
per_month(5)
```



```
per_month(10)
```



```
k = [1,2,3,4,5,6,7,8,9,10,11,12]
s1 = df[df['origin']=='EWR']['month'].value_counts().sort_index()
s2 = df[df['origin']=='LGA']['month'].value_counts().sort_index()
s3 = df[df['origin']=='JFK']['month'].value_counts().sort_index()

plt.plot(k,s1, color='orange',label='EWR')
plt.plot(k,s2, color='navy',label='LGA')
plt.plot(k,s3, color='aqua',label='JFK')
plt.legend(['EWR','LGA','JFK'])
plt.grid(True)
plt.xlabel('Month')
plt.ylabel('Frequency')
plt.show()
```

